

TWO-STAGE AND SEQUENTIAL PROCEDURES FOR ESTIMATING THE LOCATION PARAMETER OF A NEGATIVE EXPONENTIAL DISTRIBUTION

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Summary: We study the point estimation problems for the location parameter of a negative exponential distribution when the scale parameter is completely unknown. We discuss modified two-stage and sequential estimation procedures for constructing the minimum risk and bounded risk estimators. In the case of the bounded risk estimation problem, we have proved an interesting result along the lines of Simons (1968). We have explored all the procedures very extensively from many theoretical points of view and detailed analyses have been provided.

1. INTRODUCTION

We assume that $X_1, X_2, \dots$ is a sequence of independent and identically distributed (i.i.d.) random variables with the probability density function

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$$(1.1) \quad f(x; \mu, \sigma) = \sigma^{-1} \exp\{-(x-\mu)/\sigma\} I(x > \mu)$$

where $I(\cdot)$ is the usual indicator function of $(\cdot)$ and $-\infty < \mu < \infty$, $0 < \sigma < \infty$. The distribution in (1.1) is known as the negative exponential distribution with the location and scale parameters μ and σ , respectively. We assume that μ and σ are both completely unknown.

The negative exponential distribution (1.1) is used as a statistical model in various studies related to reliability experiments and clinical trials among others. One may look at Bain (1978) and Zelen (1966).

In this paper, we consider estimating μ pointwise with two different loss structures treated separately in two different sections. Having recorded $X_1, \dots, X_n$, we write $X_{n(1)} = \min(X_1, \dots, X_n)$ and

$\sigma_n = (n-1)^{-1} \sum_{i=1}^n (X_i - X_{n(1)})$ for $n \geq 2$. The statistics $X_{n(1)}$ and σ_n are used to estimate μ and σ , respectively.

The first problem we treat in this paper is the following. Suppose that the loss incurred in estimating μ by $X_{n(1)}$ is given by

$$(1.2) \quad L_n = A(X_{n(1)} - \mu)^s + cn^t$$

where A , s , c and t are all known positive numbers. The risk function associated with (1.2) is given by

$$(1.3) \quad R_n(c) = E(L_n) \\ = (K\sigma^s/s)n^{-s} + cn^t$$

where $K = As\Gamma(s+1)$. Now, the goal is to minimize $R_n(c)$, and the associated optimal fixed sample size had σ been known is given by $n^* = \{K\sigma^s(ct)^{-1}\}^{1/(s+t)}$. However, n^* is unknown. Sequential procedures were proposed in Basu (1971) and Mukhopadhyay (1974), and various first-order asymptotic (as $c \rightarrow 0$) properties were derived. Mukhopadhyay (1982a)

also proved various second-order asymptotic properties of the above-mentioned sequential procedures. In section 2, we propose the modified two-stage procedure for this problem along the lines of Mukhopadhyay (1980), and Ghosh and Mukhopadhyay (1976, 1981). We present several interesting properties of the suggested sampling technique. We also give a few remarks in connection with the moderate sample size performances of the proposed sampling rule.

Now we turn to the next problem. Suppose that the loss incurred in estimating μ by $X_{n(1)}$ is given by

$$(1.4) \quad L_n^* = A(X_{n(1)} - \mu)^s$$

where A and s are known positive numbers. The associated risk function is

$$(1.5) \quad R_n^*(A) = E(L_n^*) \\ = (K\sigma^s/s)n^{-s},$$

and our goal is to make this risk not to exceed a pre-assigned level $W(>0)$. Had σ been known, the optimal fixed sample size required to meet this goal would turn out to be the smallest integer $\geq D$ where $D = \{K\sigma^s(sW)^{-1}\}^{1/s}$. However, D is unknown. Thus, we propose the modified two-stage and sequential procedures for this specific problem in section 3. We develop many interesting properties of the suggested sampling techniques. In particular, a result has been proved in the spirit of Simons (1968). A few remarks have also been added in connection with the moderate sample size performances of the proposed sampling procedures.

2. MINIMUM RISK ESTIMATION

We may recall that we are dealing with the loss function (1.2) having the associated risk function $R_n(c)$ given in (1.3). $R_n(c)$ is minimized at $n = n^* = \{K\sigma^s(ct)^{-1}\}^{1/(s+t)}$ with the corresponding minimum risk

$R_{n^*}(c) = c(1+ts^{-1})(n^*)^t$. Since n^* is indeed unknown, we propose a modified two-stage procedure to estimate μ with the risk approximately being at the minimum. We write $[x]$ for the largest integer smaller than x .

2.1 A modified two-stage procedure

The modified two-stage procedure is proposed along the lines of Mukhopadhyay (1980, 1982b) and Ghosh and Mukhopadhyay (1981).

Let $\gamma(>0)$ be chosen and fixed in advance and we define

$$m = \max\{[1+s^2/(s+t)] + 1, [B^{1/(s+t+\gamma)}] + 1\}$$

with $B = K(ct)^{-1}$. The problem of "good" choices of γ will be discussed later. Note that we have achieved $\lim_{c \rightarrow 0} m = \infty$ and at the same time we have

$\lim_{c \rightarrow 0} (m/n^*) = 0$ by introducing the "design parameter" γ while defining m , the starting sample size. We start with $X_1, \dots, X_m$ and let

$$(2.1) \quad N = \max\{m, [B\sigma_m^2]^{1/(s+t)} + 1\},$$

and then we proceed to consider $X_{N(1)}$ as the estimator of μ . Some of the interesting properties of the modified two-stage procedure (2.1) are stated in the following theorem.

Theorem 1 For the sampling plan defined in (2.1), we have

(a) $E(N) < \infty$,

(b) $\lim_{c \rightarrow 0} (N/n^*) = 1$ a.s.,

(c) $E(N^{\omega} n^{*-\omega}) = 1 + O(c^\eta)$ as $c \rightarrow 0$ for every fixed non-zero ω where
 $\eta = \min\{(s+t+\gamma)^{-1}, (s+t)^{-1} - (s+t+\gamma)^{-1}\}$.

Let us prove first a result giving the asymptotic behaviour of some complicated gamma functions.

Lemma 1 For every fixed real number b , we have

$$\Gamma(a+b)a^{-b}\{\Gamma(a)\}^{-1} = 1 + O(a^{-1})$$

as $a \rightarrow \infty$.

Proof of Lemma 1 The proof follows immediately from the well-known logarithmic expansion which can be found, e.g., on page 130 of Cramér (1963):

$$\ln \Gamma(p) = (p - \frac{1}{2}) \ln(p) - p + \frac{1}{2} \ln(2\pi) + \tilde{R}(p),$$

where

$$\tilde{R}(p) = \frac{p}{12} - \frac{p^3}{360} + O(p^{-5}).$$

Further details are omitted.

Proof of theorem 1 We omit the proofs of parts (a) and (b) for brevity.

In order to prove part (c), we first consider $\omega > 0$.

We start with

$$N \leq \left(\frac{K \sigma_m^s}{ct} \right)^{\frac{1}{s+t}} + m,$$

and thus we have

$$\begin{aligned} (2.2) \quad E(N/n^*)^\omega &\leq E \left\{ n^{*-1} \left(\left(\frac{K \sigma_m^s}{ct} \right)^{\frac{1}{s+t}} + m \right) \right\}^\omega \\ &= E \left\{ (\sigma_m/\sigma)^{\frac{s}{s+t}} + (m/n^*) \right\}^\omega \end{aligned}$$

$$\begin{aligned}
&\leq E\left\{\left(\sigma_m/\sigma\right)^{\frac{\omega s}{s+t}}\left(1+\left(m/n^*\right)\left(\sigma_m/\sigma\right)^{-\frac{s}{s+t}}\right)^{[\omega]+1}\right\} \\
&= E\left\{\left(\sigma_m/\sigma\right)^{\frac{\omega s}{s+t}}\sum_{n=0}^{[\omega]+1}\binom{[\omega]+1}{n}\left(m/n^*\right)^n\left(\sigma_m/\sigma\right)^{-\frac{ns}{s+t}}\right\} \\
&= E\left\{\left(\sigma_m/\sigma\right)^{\frac{\omega s}{s+t}}\left(1+\sum_{n=1}^{[\omega]+1}\binom{[\omega]+1}{n}\left(m/n^*\right)^n\left(\sigma_m/\sigma\right)^{-\frac{ns}{s+t}}\right)\right\} \\
&\leq \Gamma\left(m-1+\frac{s\omega}{s+t}\right)(m-1)^{-\frac{s\omega}{s+t}}\left\{\Gamma(m-1)\right\}^{-1} \\
&\quad + K^*c^{\frac{1}{s+t}-\frac{1}{s+t+\gamma}}\sum_{n=1}^{[\omega]+1}\binom{[\omega]+1}{n}E\left\{\left(\sigma_m/\sigma\right)^{\frac{s(\omega-n)}{s+t}}\right\},
\end{aligned}$$

for c small enough so that $m = \left[\left(\frac{K}{c^t}\right)^{\frac{1}{s+t+\gamma}}\right] + 1$, and K^* a positive generic constant independent of c . In the last step leading to (2.2) we note the relationships between m and c , n^* and c , and the fact

that by definition, $m \geq \left[\frac{s^2}{s+t} + 1\right] + 1$ guarantee the existence of

$E\left\{\left(\sigma_m/\sigma\right)^{\frac{s(\omega-n)}{s+t}}\right\}$ for $1 \leq n \leq [\omega] + 1$. Applying lemma 1 to the first part of (2.2) with $a = m - 1$ and $b = s\omega/(s+t)$, and again noting the relationship between m and c , yield

$$\begin{aligned}
(2.3) \quad E(N/n^*)^\omega &\leq 1 + O\left(c^{\frac{1}{s+t+\gamma}}\right) \\
&\quad + K^*c^{\frac{1}{s+t}-\frac{1}{s+t+\gamma}}\sum_{n=1}^{[\omega]+1}\binom{[\omega]+1}{n}E\left\{\left(\sigma_m/\sigma\right)^{\frac{s(\omega-n)}{s+t}}\right\} \\
&\leq 1 + O\left(c^{\frac{1}{s+t+\gamma}}\right) + K^*c^{\frac{1}{s+t}-\frac{1}{s+t+\gamma}} \\
&\quad \times \sum_{n=1}^{[\omega]+1}\binom{[\omega]+1}{n}\Gamma\left(m-1+\frac{s(\omega-n)}{s+t}\right)(m-1)^{-\frac{s(\omega-n)}{s+t}}\left\{\Gamma(m-1)\right\}^{-1},
\end{aligned}$$

as $c \rightarrow 0$. Lemma 1 implies the boundedness of

$$\sum_{n=1}^{[\omega]+1} \binom{[\omega]+1}{n} \Gamma\left(m-1 + \frac{s(\omega-n)}{s+t}\right) (m-1)^{\frac{-s(\omega-n)}{s+t}} \left\{ \Gamma(m-1) \right\}^{-1} \quad \text{as } c \rightarrow 0, \text{ which together}$$

with (2.3) gives

$$\begin{aligned} \limsup_{c \rightarrow 0} E(N/n^*)^\omega &\leq 1 + O\left(\frac{1}{c^{s+t+\gamma}}\right) + O\left(\frac{1}{c^{s+t}} - \frac{1}{s+t+\gamma}\right) \\ &= 1 + O(c^\eta), \end{aligned}$$

$$\text{where } \eta = \min\left\{\frac{1}{s+t+\gamma}, \frac{1}{s+t} - \frac{1}{s+t+\gamma}\right\}.$$

Also, by definition of N ,

$$N \geq \left(\frac{K \sigma_m^s}{c t}\right) \frac{1}{s+t},$$

which implies that

$$(N/n^*)^\omega \geq (\sigma_m/\sigma)^{\frac{s\omega}{s+t}}.$$

Thus,

$$\begin{aligned} \liminf_{c \rightarrow 0} E(N/n^*)^\omega &\geq \liminf_{c \rightarrow 0} E\left\{(\sigma_m/\sigma)^{\frac{s\omega}{s+t}}\right\} \\ &= \liminf_{c \rightarrow 0} \Gamma\left(m-1 + \frac{s\omega}{s+t}\right) (m-1)^{\frac{-s\omega}{s+t}} \left\{ \Gamma(m-1) \right\}^{-1} \\ &= 1 + O\left(\frac{1}{c^{s+t+\gamma}}\right), \end{aligned}$$

using lemma 1. Putting together the $\limsup$ and $\liminf$ parts yields

$$(2.4) \quad \lim_{c \rightarrow 0} E(N/n^*)^\omega = 1 + O(c^\eta),$$

for positive ω .

Let us now turn to the behaviours of the negative moments of N/n^* . Thus, suppose that ω is a fixed negative number. We start with

$$N \geq \left(\frac{K \sigma_m^s}{ct} \right)^{\frac{1}{s+t}}$$

which leads to

$$\begin{aligned} E(N/n^*)^{-\omega} &\leq E \left\{ n^{*-1} \left(\frac{K \sigma_m^s}{ct} \right)^{\frac{1}{s+t}} \right\}^{-\omega} \\ &= E \left\{ (\sigma_m/\sigma)^{\frac{s\omega}{s+t}} \right\} \\ &= \Gamma(m-1-s\omega/(s+t)) (m-1)^{\frac{s\omega}{s+t}} \left\{ \Gamma(m-1) \right\}^{-1}, \end{aligned}$$

for c small enough so that $m > 1 + s\omega/(s+t)$. Therefore,

$$\limsup_{c \rightarrow 0} E(N/n^*)^{-\omega} \leq 1 + O \left(c^{\frac{1}{s+t+\gamma}} \right),$$

using lemma 1. Now,

$$\begin{aligned} E(N/n^*)^{-\omega} &= E \left\{ ((N/n^*)^\omega)^{-1} \right\} \\ &\geq \left\{ E(N/n^*)^\omega \right\}^{-1}, \end{aligned}$$

by Jensen's inequality. Thus, using (2.4), we have

$$\begin{aligned} \liminf_{c \rightarrow 0} E \left\{ (N/n^*)^{-\omega} \right\} &\geq \left\{ 1 + O(c^\eta) \right\}^{-1} \\ &= 1 + O(c^\eta). \end{aligned}$$

Putting the $\limsup$ and $\liminf$ parts together yields

$$\lim_{c \rightarrow 0} E(N/n^*)^{-\omega} = 1 + O(c^\eta),$$

which completes the proof of theorem 1.

Let us now compute the risk associated with the procedure (2.1). First note that the event " $N = n$ " is independent of $X_{n(1)}$ for every fixed $n = m, m+1, \dots$. Thus, the risk associated with the estimator $X_{N(1)}$ is given by

$$(2.5) \quad R(c) = E(L_N) \\ = (K\sigma^2/s)E(N^{-s}) + cE(N^t).$$

For simplicity, we will write R_{n^*} and R instead of $R_{n^*}(c)$ and $R(c)$ respectively.

Theorem 2 For the sampling plan defined in (2.1) we have

- (a) $\lim_{c \rightarrow 0} R/R_{n^*} = 1,$
 (b) $R - R_{n^*} = O(c^{\frac{s}{s+t} + \eta})$ as $c \rightarrow 0$, where
 $\eta = \min\{(s+t+\gamma)^{-1}, (s+t)^{-1} - (s+t+\gamma)^{-1}\}.$

Proof First note that

$$R/R_{n^*} = (1+t/s)^{-1} \{E\{(N/n^*)^t\} \\ + (t/s)E\{(N/n^*)^{-s}\}\}.$$

In order to prove part (a), we now use part (c) of theorem 1 with $\omega = t$ and $\omega = -s$. One may note that the restrictions put on m are actually needed in these verifications.

To verify part (b), we again use part (c) from theorem 1 with $\omega = t$ and $\omega = -s$, and we write, for small c ,

$$(R - R_{n^*}) = c^{1 - \frac{t}{s+t}} (K\sigma^2/t)^{\frac{t}{s+t}} \{(t/s)E\{(N/n^*)^{-s}\} + E\{(N/n^*)^t\} - (1+t/s)\} \\ = (K\sigma^2/t)^{\frac{t}{s+t}} c^{\frac{s}{s+t}} \{(t/s)(1+O(c^\eta)) + 1 + O(c^\eta) - (1+t/s)\},$$

and hence we obtain

$$R - R_{n^*} = O\left(c^{\frac{s}{s+t} + \eta}\right)$$

as $c \rightarrow 0$. This completes the proof of theorem 2.

Remark 1 The order of the "regret" function $R - R_{n^*}$ has been found to be $O\left(c^{\frac{s}{s+t} + \eta}\right)$ as $c \rightarrow 0$ for the modified two-stage procedure (2.1). We conjecture, however, that the order in part (b) of theorem 2 would be $O\left(c^{\frac{1}{s+t} + \frac{1}{s+t+\gamma}}\right)$ as $c \rightarrow 0$.

2.2 Some comments on the moderate sample size performances

In order to study the moderate sample size behaviours of our procedure (2.1), we have used simulations for various parameter values. We used $A = 50$, $\sigma = 1$, $\mu = 10$, $s = 2$, $t = 1$ and $n^* = 10, 15, 20, 25, 50, 100$. Then, we had $c = K\sigma^s(tn^*s+t)^{-1}$. We implemented the modified two-stage procedure (2.1) with $\gamma = 0.2, 0.3$. For each value of n^* , we repeated the simulated experiment 200 times in order to estimate $E(N)$, $E(L_N)$ together with their standard errors. The simulations were run on an IBM 3081 computer using the SAS language, and the SAS (1982) function RANEXP was utilized to generate the pseudo-random negative exponential variates.

The results of the simulations were encouraging. The average sample size could be seen to closely approximate n^* . Our choices of γ while implementing the modified two-stage procedure (2.1) were based largely on simulations of other procedures which have been carried out in the past for other related problems. We find $\gamma = 0.2$ or 0.3 as a "good" choice. We prefer not to give any of our tables here to save some space.

BOUNDED RISK ESTIMATION

We should recall that we are dealing with the loss function (1.4) having the associated risk function $R_n^*(A)$ given in (1.5). Given a preassigned number $W(>0)$, the goal that the associated risk does not exceed W is achieved by taking a fixed sample of optimal size $D = \{K\sigma^S(sW)^{-1}\}^{1/S}$, had σ been known. However, D is unknown since σ is unknown, and thus we take resort to implementing various two-stage and sequential procedures such that our goal is met exactly or asymptotically (as $A \rightarrow \infty$). We continue writing $[x]$ for the largest integer smaller than x .

3.1 A modified two-stage procedure

We now propose a modified two-stage procedure for this problem along the lines of Mukhopadhyay (1980, 1982b) and Ghosh and Mukhopadhyay (1981). Let $\gamma(>0)$ be chosen and fixed in advance and we define (with $B^* = K(sW)^{-1}$)

$$m = \max\{[1+s] + 1, [(B^*)^{1/(s+\gamma)}] + 1\}.$$

Obviously, $m \rightarrow \infty$ as $A \rightarrow \infty$, however, $m/D \rightarrow 0$ as $A \rightarrow \infty$. We start with $X_1, \dots, X_m$ and let $h = (m-1)^S \{\Gamma(m-1)\}^{-1} \Gamma(m-1-s)$, and

$$(3.1) \quad N = \max\{m, [(B^* h \sigma_m^S)^{1/S}] + 1\}.$$

Then, we proceed to consider $X_{N(1)}$ as the estimator of μ . Some of the interesting properties of the modified two-stage procedure (3.1) are listed in the following theorem.

Theorem 3 For the stopping variable N defined in (3.1), we have

- (a) $E(N) < \infty$,
- (b) $\lim_{A \rightarrow \infty} (N/D) = 1$ a.s.,
- (c) $\lim_{A \rightarrow \infty} E(N/D) = 1$,

$$(d) \quad R^* = E(L_N) \leq W.$$

Proof of theorem 3 is omitted for the sake of brevity.

We now state and prove the following lemma.

Lemma 2 For every real number m , we have:

$$(a) \quad (m-1)^{-\rho} \Gamma(m-1+\rho) \{\Gamma(m-1)\}^{-1} < 1 \quad \text{if } 0 < \rho < 1 \quad \text{and } m > 1,$$

$$(b) \quad (m-1)^{\rho} \Gamma(m-1-\rho) \{\Gamma(m-1)\}^{-1} > 1 \quad \text{if } \rho > 0 \quad \text{and } m > 1 + \rho.$$

Proof of Lemma 2 To prove part (a), let $G = (m-1)^{-\rho} \Gamma(m-1+\rho)$ and $H = \Gamma(m-1)$. Now,

$$\begin{aligned} G &= (m-1)^{-\rho} \int_0^{\infty} x^{(m-1+\rho)-1} e^{-x} dx \\ &= (m-1)^{-\rho} \Gamma(m-1) E(X^{\rho}) \end{aligned}$$

where $X \sim \text{Gamma}(1, m-1)$. Now, $h(x) = x^{\rho}$ is a concave function for $x > 0$ and $0 < \rho < 1$. Thus, using Jensen's inequality, we can easily see that

$$\begin{aligned} G - H &= (m-1)^{-\rho} \Gamma(m-1) \{E(X^{\rho}) - (m-1)^{\rho}\} \\ &< (m-1)^{-\rho} \Gamma(m-1) \{(EX)^{\rho} - (m-1)^{\rho}\}, \end{aligned}$$

and this leads to part (a). The proof of part (b) is similar, noting that $h(x) = x^{-\rho}$ is a convex function for $\rho > 0$ and $x > 0$.

Remark 2 Using part (b) of lemma 2 and lemma 1, one can show that $E(N-D)$ explodes in the limit as $A \rightarrow \infty$, where N comes from (3.1). Thus, we go for a purely sequential procedure in order to be able to conclude that $E(N-D)$ is bounded in the limit as $A \rightarrow \infty$ for that procedure.

3.2. A sequential procedure

We start the experiment with $m (\geq 2)$ samples, and then we take one sample at a time, if necessary, according to the following stopping rule. Let

$$(3.2) \quad N = \inf\{n \geq m: n \geq (B^* \sigma_n^S)^{1/s}\}.$$

It can be shown that $P(N < \infty) = 1$, and thus when the rule (3.2) stops, we estimate μ by $\bar{X}_{N(1)}$.

Theorem 4 For the stopping variable N defined in (3.2), we have

- (a) $E(N) < \infty$,
- (b) $\lim_{A \rightarrow \infty} (N/D) = 1$ a.s.,
- (c) $P\{N \leq (1-\epsilon)D\} = O_e(D^{-(m-1)})$ as $A \rightarrow \infty$ for every fixed ϵ in $(0,1)$,
- (d) $\lim_{A \rightarrow \infty} E(N/D) = 1$.

Proof of theorem 4 follows along the lines of Mukhopadhyay (1974). Further details are omitted.

Before we state the next few results, let us first look at an alternative representation of stopping rule (3.2). Suppose $Y_1, Y_2, \dots$ is a sequence i.i.d. random variables having the probability density function $f(y; 0, \sigma)$ as defined in (1.1). Let us define

$$(3.3) \quad N^* = \inf\left\{n \geq m: \sum_{i=1}^{n-1} Y_i \leq n(n-1)B^{*-1/s}\right\},$$

and then following the techniques of Lombard and Swanepoel (1978) and Swanepoel and van Wyk (1982), we can show that the two stopping times defined in (3.2) and (3.3) are indeed identically distributed. Furthermore, if we write

$$(3.4) \quad T = \inf\{n \geq m-1: \sum_{i=1}^n Y_i \leq B^{*-1/s} n^2(1+n^{-1})\},$$

one can see that N , or equivalently N^* , equals $T + 1$ and T has the same form as in (1.1) of Woodroffe (1977). Let us write

$$\nu = \sigma - \sum_{n=1}^{\infty} n^{-1} E \left\{ \left(\sum_{i=1}^n Y_i - 2n\sigma \right)^+ \right\}$$

where $x^+ = \max(0, x)$. Let $\eta = \nu\sigma^{-1} - 1$. From table 2.1 in Woodroffe (1977) one can see that $\nu = 0.747\sigma$. Now, we state the following theorem.

Theorem 5 For the stopping variable N defined in (3.2), we have

as $A \rightarrow \infty$:

(a) $E(N) = D + \eta + o(1)$ if $m \geq 3$,

(b) The associated risk is given by

$$\begin{aligned} E(L_N^*) &= \infty && \text{if } m < s+1, \\ &= W + Ws^{s-1} \{\Gamma(s)\}^{-1} && \text{if } m = s+1, \\ &= W + O(A^{(s-m+1)/s}) && \text{if } s+1 < m < s+2 \text{ and } m \geq 3, \\ &= W + O(A^{-1/s}) && \text{if } m \geq s+2. \end{aligned}$$

Proof Part (a) follows from the fact that $E(T) = D + \nu\sigma^{-1} - 2 + o(1)$ which is easy to verify utilizing Woodroffe's (1977) theorem 2.4.

Now, turning to part (b), one can see that the two cases $m < s + 1$ and $m = s + 1$ can be treated as in Starr (1966) and Mukhopadhyay (1974). In order to verify the remaining two cases, one may use techniques similar to those used in Mukhopadhyay (1982a), and in doing so we exploit the representation (3.4) and Woodroffe's second-order expansions. All further details are omitted for brevity.

Remark 3 Our part (a) in theorem 5 proves that the purely sequential procedure (3.2) is "asymptotically second-order efficient" in the Ghosh-Mukhopadhyay (1981) sense. However, by combining part (c) of theorem 3 and the first part of remark 2, one can see that the modified two-stage procedure (3.1) is only "asymptotically first-order efficient" in the

Ghosh-Mukhopadhyay (1981) sense.

Remark 4 Let us explain an interesting point. One can see from theorem 5 that $E(L_N^*) \rightarrow W$ as $A \rightarrow \infty$ when $m \geq s + 2$. However, we started section 3 with the goal in mind that we needed $E(L_N^*) \leq W$. Now, the sequential procedure (3.2) cannot quite guarantee that. However, we can prove an existential result which looks almost like this. We can show that there exists a nonnegative integer P , which is independent of A , σ , μ and W , such that by taking P extra samples after the rule (3.2) stops, we can guarantee $E(L_{N+P}^*) \leq W$ for $m \geq s + 2$. Simons (1968) proved such a result in a different context though.

We now state the following lemma without giving its proof.

Lemma 3 Let P be a nonnegative real number. If we expand $(N^*+P)^{-s}$ in a Taylor series, we obtain

$$(N^*+P)^{-s} = D^{-s} - sD^{-(s+1)}(N^*+P-D) + \frac{1}{2}s(s+1)Q^{-(s+2)}(N^*+P-D)^2$$

where $Q^{-(s+2)} \leq K^*D^{-2}$, K^* being a generic constant independent of A , σ , μ and W .

Theorem 6 There exists a nonnegative integer P independent of A , σ , μ and W such that

$$E(L_{N+P}^*) \leq W,$$

if $m \geq s + 2$ where N is the stopping variable defined by (3.2).

Proof First note that the distribution of N , and hence that of N^* also, depends only on D . From lemma 3 we know that

$$(N^*+P)^{-s} = D^{-s} - sD^{-(s+1)}(N^*+P-D) + \frac{1}{2}s(s+1)Q^{-(s+2)}(N^*+P-D)^2,$$

where Q is between $N^* + P$ and D . Thus,

$$(3.5) \quad E(L_{N^*+P}^*) = Wn^*s\{D^{-s}-sD^{-(s+1)}\}E(N^*+P-D) \\ + \frac{1}{2}s(s+1)\{E\{Q^{-(s+2)}(N^*-D)^2\} + 2PE\{Q^{-(s+2)}(N^*-D)\} \\ + P^2E\{Q^{-(s+2)}\}\}.$$

Combining (3.5) with part (a) of theorem 5, we obtain

$$(3.6) \quad E(L_{N^*+P}^*) = W\{1-sD^{-1}(P+\beta+o(1)) + \frac{1}{2}s(s+1)D^2 \\ \times \{E\{Q^{-(s+2)}(N^*-D)^2I(N^*+1 \leq (1-\varepsilon)D)\} \\ + E\{Q^{-(s+2)}(N^*-D)^2I(N^*+1 > (1-\varepsilon)D)\} \\ + 2PE\{Q^{-(s+2)}(N^*-D)I(N^* \leq D)\} \\ + 2PE\{Q^{-(s+2)}(N^*-D)I(N^* > D)\} \\ + P^2E\{Q^{-(s+2)}I(N^*+1 \leq (1-\varepsilon)D)\} \\ + P^2E\{Q^{-(s+2)}I(N^*+1 > (1-\varepsilon)D)\}\},$$

as $D \rightarrow \infty$ for ε in $(0,1)$. Making various simplifications, again noting that there exists n_0 such that $D > n_0$ and $N^* + 1 > (1-\varepsilon)D$ will imply that $Q^{-(s+2)} \leq K^*D^{-(s+2)}$, (3.6) becomes

$$(3.7) \quad E(L_{N^*+P}^*) \leq W\{1-sD^{-1}(P+\beta+o(1)) + kD^s \\ \times \{E\{Q^{-(s+2)}(N^*-D)^2I(N^*+1 \leq (1-\varepsilon)D)\} \\ + K^*D^{-(s+1)}E\{(N^*-D)^2/D\} \\ + 2PD^{-(s+3/2)}E\{|N^*-D|/D^{1/2}\} \\ + P^2E\{Q^{-(s+2)}I(N^*+1 \leq (1-\varepsilon)D)\} \\ + K^*P^2D^{-(s+2)}P(N^*+1 > (1-\varepsilon)D)\} \\ \leq W\{1-sD^{-1}(P+\beta+o(1)) \\ + K^*D^sE\{Q^{-(s+2)}(N^*-D)^2I(N^*+1 \leq (1-\varepsilon)D)\}$$

$$\begin{aligned}
 &+ K^* D^{-1} O(1) + 2P O(D^{-3/2}) \\
 &+ P^2 D^S E\{Q^{-(s+2)} I(N^*+1 \leq (1-\varepsilon)D)\} \\
 &+ K^* P^2 D^{-2} \}
 \end{aligned}$$

as $D \rightarrow \infty$, where throughout K^* is a generic constant independent of μ , σ , and W . Now, using lemma 3 on the two terms in (3.7) which remain unevaluated, and noting the relationship between N and N^* , we have

$$\begin{aligned}
 (3.8) \quad E(L_{N^*+p}^*) &\leq W\{1 - sD^{-1}(P + \beta + O(1)) \\
 &\quad + K^* D^S E\{(N^* - D)^2 D^{-2} I(N^*+1 \leq (1-\varepsilon)D)\} \\
 &\quad + O(D^{-1}) + P O(D^{-3/2}) \\
 &\quad + K^* P^2 D^{S-2} E\{I(N^*+1 \leq (1-\varepsilon)D) + P^2 O(D^{-2})\} \\
 &\leq W\{1 - sD^{-1}(P + \beta + O(1)) + K^* D^S P\{N \leq (1-\varepsilon)D\} \\
 &\quad + O(D^{-1}) + P O(D^{-3/2}) + K^* P^2 D^{S-2} P\{N \leq (1-\varepsilon)D\} + P^2 O(D^{-2})\}
 \end{aligned}$$

as $D \rightarrow \infty$. Now, using theorem 4 part (c), (3.8) becomes

$$\begin{aligned}
 (3.9) \quad E(L_{N^*+p}^*) &\leq W\{1 - sD^{-1}(P + \beta + O(1)) + O(D^{S-m+1}) \\
 &\quad + O(D^{-1}) + P O(D^{-3/2}) + P^2 O(D^{S-m-1}) \\
 &\quad + P^2 O(D^{-2})\} ,
 \end{aligned}$$

as $D \rightarrow \infty$. Simplifying (3.9) further, with the assumption $m \geq s + 2$, yields

$$E(L_{N^*+p}^*) \leq W\{1 - sD^{-1}(P + \beta + O(1)) + O(D^{-1}) + P O(D^{-3/2}) + P^2 O(D^{-2})\}$$

as $D \rightarrow \infty$. Thus, $E(L_{N^*+p}^*) \leq W$ if

$$sD^{-1}(P + \beta + O(1)) \geq O(D^{-1}) + P O(D^{-3/2}) + P^2 O(D^{-2}) ,$$

or

$$P + \beta + o(1) \geq O(1) + PO(D^{-\frac{1}{2}}) + P^2O(D^{-1}),$$

or

$$(3.10) \quad P + \beta \geq O(1) + PO(D^{-\frac{1}{2}}) + P^2O(D^{-1})$$

as $D \rightarrow \infty$.

Now, (3.10) is satisfied for all large D , say $D > n_0$, and for some large P , say $P \geq P_0$. If $D \leq n_0$, then take $P \geq P_1$ and we will have $E(L_{N^*+P_1}^*) \leq W$. Thus, for $P^* \geq \max\{P_0, P_1\}$ we have $E(L_{N^*+P^*}^*) \leq W$. Noting that N has the same probability distribution as N^* , and that L_n is decreasing in n , we have $E(L_{N+P^*}) \leq W$. Thus, choosing $P \geq P^*$ completes the proof.

3.3 Some comments on the moderate sample size performances

In order to study the moderate sample size behaviours of the procedures introduced in this section, we considered $W = 1$ while the choices for s , μ , σ , m , γ and D (like n^*) remained the same as they were in section 2.2. We computed A from the relation $D^S = K\sigma^S(sW)^{-1}$. Again we used only 200 repetitions to estimate $E(N)$ and $E(L_N^*)$ together with their standard errors.

In the case of the modified two-stage procedure where we do have theoretically $R^* \leq W$, we have very satisfactory moderate sample size behaviours, but at the expense of slight oversampling. The modified two-stage procedure seems to behave extremely well with only a few extra samples compared to D .

The results were particularly not very satisfying in the sequential case. Inadequacy of 200 repetitions, a restriction put forth by funding constraints, were very apparent. Our particular simulated tables were quite unstable in this case. However, the simulated findings of Başı (1971),

Mukhopadhyay (1974), and Swanepoel and van Wyk (1984) would give us enough confidence in recommending any of these procedures of section 3 without any hesitation. Again, we omit all our tables for the sake of brevity.

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